10 · Project sheet · Ayo's Bread Delivery

Robot / Maker Lab practical · Nursery 1 · Term 3 · Week 9 · Integration

1. At a glance

Time	40 minutes (fits the lesson; 20 min if used on its own)
Group size	Teams of 4 (planner, Ayo, checker, light-holder)
Learning goal	I can use my thinking steps to solve Ayo's problem.
Coding idea	Use the thinking steps together: problem → small steps → safe way → if–then → another way.
Week link	https://rhinoxscientifica.com/q/n1-t3-w9

2. Materials needed

Item	Per	Low-cost swap
Taped grid + box bread shop + Grandma's house sign	class	Chalk squares; chairs with signs
Road-closed box	class	A cushion
Red and green cards	team	Coloured paper
Step cards incl. go round	team	Hand signs
Role cards	team	Say the jobs
Cardboard Ayo + paper bread	team	A toy and a paper ball

3. Safety and care

- Walk on the grid.
- Boxes light and empty.
- One team on the grid at a time.
- Stop on red; cross real roads with a grown-up.

4. Before the lesson

1. Set up the grid with shop and Grandma's house.
2. Print cards, role cards and the thinking-steps poster.
3. Read the lesson spec the evening before and lay out the cards.

5. Step-by-step guide

Step	Teacher says	Children do
1	"What is Ayo's problem?"	"Grandma needs bread!"
2	"Planner: lay 3 small steps."	Pick up → walk safe → give
3	"Ayo, walk the plan. Checker, point."	Walk the route
4	(Box + red card) "Red and blocked!"	Stop!

Step	Teacher says	Children do
5	“What now?”	Swap in “go round”; green → go
6	“Here’s your bread, Grandma!”	Deliver; photo the plan

6. Explore it

Question	Can we use all our thinking steps on one new problem?
Predict	“What will we do if the road closes?”
Try	Plan, walk, and change the plan when blocked.
Observe	“Which thinking steps did we use?”
Explain	Thinkers use their steps together to solve a new problem.

7. Talk about it

1a. What was the problem?

1b. What were your 3 small steps?

1c. What did you do on red?

2a. Which card did you change?

2b. What was your job?

2c. How did Ayo get the bread to Grandma?

8. Support and stretch

Support	The teacher asks each thinking step as a question, one at a time.
Stretch (optional)	A second closed road, or a 4-step delivery. Never assessed.

9. How we check

Observation only, not marked. Photo each plan.

Look for	Secure	With prompts	Not yet
Says the problem	☐	☐	☐
Plans 3 small steps	☐	☐	☐
Stops on red	☐	☐	☐
Finds another way when blocked	☐	☐	☐

10. Take-home and parent note

> **Dear parent or carer**
>
> Today your child was a **thinker**: their team planned Ayo’s bread delivery and changed the plan when the road closed.
>
> At home: plan a little “delivery” in 3 steps together.

11. Show and share

- Teams show their plan photos and act the surprise moment.
- Everyone says: “I am a thinker!”
- Every child who used the thinking steps gets a **Step Star sticker**.



The Step Star is a weekly sticker, not a badge. The term badge “Step Thinker · Term 3” is awarded in Week 13.